Journal of Science and Medicine in Sport 29 (2026) 709–715

Contents lists available at ScienceDirect

Journal of Science and Medicine in Sport

journal homepage: www.elsevier.com/locate/jsams

Original research

Knowledge of sport-related concussion among referees in youth football in Croatia, Germany, Norway, and Türkiye – a survey-based study

Giulia Gorgoni [a,*], Gabriel Monthuley [a], Thor Einar Andersen [b,c,d], Nora Kümmeth-Gangel [e], Ivana Pavlinac [f], Zoran Dogas [f], Anna Nordström [g,h]

[a] Institute of Sports and Preventive Medicine, Saarland University, Germany
[b] The Oslo Sports Trauma Research Centre, Department of Sports Medicine, The Norwegian School of Sport Sciences, Norway
[c] The Norwegian Football Association's Sports Medicine Clinic, Norway
[d] The Football Association of, Norway
[e] German Football Association, Germany
[f] Department for Neuroscience, School of Medicine, Sleep Medicine Center, University of Split, Croatia
[g] Department of Medical Sciences, Rehabilitation Medicine, Uppsala University, Sweden
[h] School of Sport Sciences, UiT The Arctic University of Norway, Norway

ARTICLE INFO

Article history:
Received 1 October 2025
Received in revised form 14 April 2026
Accepted 19 April 2026
Available online 28 April 2026

Keywords:
Brain injuries
Health education
Cross-cultural comparison
Injury prevention
Professional role

ABSTRACT

Objectives: Sport-related concussions (SRCs), though common in football, remain under-recognized, particularly at youth and amateur levels where medical personnel are rarely present. Referees are key stakeholders in safeguarding player health, yet their knowledge and awareness of SRCs are underexplored. This study assessed referees' knowledge and awareness about SRCs across Europe.
Design: Cross-sectional online survey.
Methods: A 19-item questionnaire was distributed by national football associations in Croatia, Germany, Norway, and Türkiye. Items addressed demographics, concussion knowledge, symptom recognition, awareness of assessment tools, and interest in education. A concussion knowledge score (CKS) was calculated. Data were analyzed using Kruskal-Wallis and Wilcoxon rank-sum test, with significance set at $p < 0.05$.
Results: 2722 surveys were analyzed. The overall median CKS was 75% with no significant differences between countries ($p = 0.345$). In Germany, CKS differed by gender and prior SRC history (significant differences, $p < 0.01$). Commonly recognized symptoms included dizziness (≥90%) and headache (≥78%), whereas subtler signs like irritability (≤21%) were poorly identified. Eleven percent of referees reported awareness of SRC assessment tools, most citing eye movement tests; only few named formal tools such as Sport Concussion Assessment Tool (SCAT) or Concussion Recognition Tool (CRT). Across all respondents, 73% expressed interest in concussion education.
Conclusions: Referees demonstrated moderate SRC knowledge but notable gaps in recognizing subtle symptoms and awareness of validated assessment tools. Due to the highly uneven sample distribution, between-country comparisons are exploratory. Incorporating concussion education into referee licensing may enhance early identification and promote player safety in youth and amateur football.
© 2026 The Author(s). Published by Elsevier Ltd on behalf of Sports Medicine Australia. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

Practical implications

- Education around concussion should emphasize behavioral/cognitive signs and symptoms alongside classic physical symptoms.
- REFORM programs will focus on training referees to systematically recognize subtle concussion signs during play and promptly remove the player(s) from the pitch or, if present, alert medical staff.
- Comprehensive and more education on formal tools (i.e., CRT6) to recognize concussions on the pitch is desired and valued by referees.

1. Introduction

Sport-related concussions (SRCs) are among the most common head injuries in football.[1,2] Epidemiological studies report match-play SRC incidence rates between 0.16 and 2.44,[3] with most concussions affecting players under the age of 25.[2] An SRC is a traumatic brain injury caused by a direct blow to the head, neck, or body, resulting in an impulsive force transmitted to the brain.[4] Symptoms may appear immediately or evolve gradually over minutes up to 48 h, complicating recognition and making diagnosis challenging.[5] Inadequate management of concussions may aggravate short-term and cause long-term neurocognitive and psychological impairments.[6,7]

* Corresponding author.
E-mail address: giulia.gorgoni@uni-saarland.de (G. Gorgoni).

https://doi.org/10.1016/j.jsams.2026.04.009
1440-2440/© 2026 The Author(s). Published by Elsevier Ltd on behalf of Sports Medicine Australia. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

As concluded by Rosenbloom et al.,[8] improvement in concussion knowledge among football referees is required to ensure the safeguarding of players with suspected concussion. Their call for action was based on a cross-sectional survey of English football referees in which 208 referees responded, yet only approximately half reported feeling confident in recognizing concussion on the pitch. Furthermore, 62% of respondents felt confident in immediate concussion management and 69% believed their knowledge to be adequate to officiate, suggesting that a substantial proportion of referees perceive limitations in their preparedness.[8] Although awareness and management of SRCs have improved in professional football in recent years,[9] and major football governing bodies, including UEFA, FIFA, and FIFPRO[10–12] have implemented targeted concussion awareness and education initiatives for key stakeholders, these knowledge gaps among referees remain. Moreover, existing research has largely examined concussion knowledge across multiple sports or in non-football populations and has predominantly focused on adult participants. Consequently, there is uncertainty regarding knowledge levels and decision-making processes among youth football referees and other non-medical stakeholders at the community level.[13] Of the limited research available, similar trends to those described by Rosenbloom et al.[8] are observed; for example, King and Coughlan[14] reported that although 85% of youth sport referees (of whom 53% were football referees) felt they could play an important role in concussion recognition, only 41% considered themselves adequately equipped with the knowledge and skills to recognize concussion during play.

At youth and grassroot levels, medical personnel are rarely present in training and/or matches, and non-medical stakeholders are often required to make critical decisions about whether a player should continue to play or be removed. In this context, referees play a central role in on-field safety, holding authority to stop play and intervene in the case of suspected injury. They act as neutral decision-makers at a critical point to decide whether a potentially concussed athlete should be medically assessed or not. Formally responsible for the player welfare during match-play, referees are a critical link in the concussion management pathway. Concussion-related knowledge directly intersects with these responsibilities. Referees must make rapid judgments under pressure, frequently from a distance, and often in the presence of subtle signs such as behavioral or cognitive changes. Thus, to improve youth football safety, it is important to assess and continuously strengthen referees' knowledge of SRCs, symptom recognition, and awareness of validated on-field tools.[13,14]

This highlights an urgent need for targeted concussion education for youth football referees.

Therefore, the primary aim of this study was to assess youth football referees' knowledge and awareness of SRCs, and of validated on-field recognition tools, including exploration of differences across subgroups (e.g., gender, self-reported history of SRCs) where sample sizes were allowed. In the long term, these findings will inform the development of targeted educational measures and standardized protocols to improve concussion management among referees in youth football, with the findings intended to serve as a structured need assessment for referee-focused concussion education within the REFORM project.

2. Methods

An ethics approval for this study was obtained from the Ethics Committee of Saarland University on 27/06/2024 (reference number: 24-21). The study forms part of the *REFORM* project (acronym for "European Football concussion awareness, prevention and management model"), funded by the European Union Erasmus + Sport (grant number: 101133951) and formally supported by the Union of European Football Associations (UEFA). The REFORM partners included UiT the Arctic University of Norway and the Football Association of Norway (NFF), the University of Split and the Croatian Football Federation (HNS), Saarland University and the German Football Association (DFB), Acibadem University and the Turkish Football Federation (TFF). All the partners approved the study.

2.1. Survey design and development

The survey was developed in English for referees affiliated with the aforementioned national football associations (NFF, HNS, DFB, and TFF). The first draft was written by a sport scientist (GG), drawing on the 6th Consensus Statement on Concussion in Sport,[4] the Football Association (FA) Concussion Guidelines, and previous work by Rosenbaum & Arnett.[15] It was first presented during an in-person meeting in Oslo on March 23rd, 2024, where at least one representative of each REFORM partner institution and each national football association was present. The questions were discussed one by one within the group; some were excluded, others categorized as "debatable" or "necessary". The questions in the category "debatable" were reassessed (GG) and subsequently discussed with two sport scientists (EV & GM) from UEFA in an online meeting, during which the wording was adjusted to optimize comprehensibility and reduce bias.

The final version, approved by the REFORM Quality Committee, comprised 19 items and is presented, in English, in Supplements. The items can be grouped into five sections: (1) demographics (a total of seven questions); (2) binary questions (three questions); (3) symptom/sign checklist (one checklist of 25 symptoms/signs and distractors); (4) awareness of specific concussion recognition/assessment tools (5 questions); and (5) interest in follow-up and additional information (3 questions).

Sections (2) and (3) were combined to calculate the concussion knowledge score (CKS). This consisted of three binary questions (true/false answers) and a checklist of 25 symptoms/signs (18 correct, seven distractors intentionally incorporated to reduce acquiescence ('yes') bias and to help identify inattentive responding). The symptom/sign answers were also treated as binary answers (correct/incorrect). Each correct answer scored one point for a total score range of 0–28. Adaptive questioning was used; some items were only shown based on prior responses to reduce survey length and complexity. Respondents were able to review and change their answers using a Back button before submitting the questionnaire.

The survey was implemented online (Netigate) and tested in English (the master version of the survey) in the REFORM working group, for its usability and technical functionality. It was then translated into Croatian, German, Norwegian, and Turkish, respectively, with both versions reviewed by native level speakers with a role in the REFORM working group, ensuring the questions were comprehensible, relevant, and free of ambiguity. Additionally, all versions were back-translated into English by a bilingual professional translator to confirm the accuracy of the translation and consistency with the master version.

Data protection was ensured by the survey platform's General Data Protection Regulation (GDPR)-compliant infrastructure, and no identifiable data was collected. Internet protocol (IP) addresses were not stored, and all responses were exported to and securely stored on Netigate's password-protected servers, accessible only to the research team.

The study was conducted and reported in accordance with the Checklist for Reporting Results of Internet E-Surveys (CHERRIES) guidelines, which are provided in the Appendix.

2.2. Participants, recruitment, and distribution

Referees, officiating up to the U23 age group, in Croatia, Germany, Norway, and Türkiye, were approached. Recruitment was facilitated and supported by the national football associations (HNS, DFB, NFF, and TFF). Each federation distributed the survey via an official email in October 2024, on a date determined independently by the association. Dissemination volume was controlled independently by each federation. The email included a cover letter outlining the study objectives, expected completion time (5–7 min), type of data to be collected, and the investigators' contact details. Participation was voluntary,

anonymous, and uncompensated. The survey link, included in the cover letter, remained open for three weeks. Data collection was conducted between November and December 2025, with each participating federation determining the exact start date internally.

2.3. Statistical analysis

Raw data were exported from Netigate (Netigate AB, Stockholm, Sweden) into Microsoft Excel (version 16.0; Microsoft Corporation, Redmond, WA, USA) for initial overview and analyses, and then analyzed in R (version 4.4.1; R Foundation for Statistical Computing, Vienna, Austria) in RStudio (version 2024.09.0 + 375; Posit, PBC, Boston, MA, USA). Data management was performed using dplyr and tidyr packages, and ggplot2 was used for data visualization. Data were excluded if participants did not answer to at least one of the three binary knowledge questions (section (2)). Descriptive statistics (median and interquartile ranges, IQR) were calculated for all quantitative variables (including age, years of refereeing experience, and the CKS), as distributions were not normal. Categorical variables (including gender, country, officiating level, history of SRC, and awareness of assessment tools) were summarized using frequencies and percentages. For item-level analyses of the symptom-recognition checklist, percentages were calculated using the number of respondents who answered each specific item (available-case analysis). This approach prevents artificially lowering of percentages due to missed items and reflects actual recognition rate among participants who provided a response. Due to partial dropout within the German sample while completing the symptom checklist ($n = 140$), denominators varied across symptom items (see Table 2); all other country samples completed the knowledge questions in full. Following Rosenbaum & Arnett,[15] correct answers were scored as 1, incorrect answers as 0, and summed to derive the concussion knowledge score (CKS; range 0–28). For participants with incomplete responses, the CKS was calculated based on the number of items completed and expressed as a percentage of correct responses, to facilitate interpretability. A Kruskal-Wallis test was used to examine differences in CKS between countries. When significant, Dunn's post hoc test with Bonferroni correction was applied for multiple pairwise comparisons. Due to limited subgroup sample sizes (e.g., for level of play, prior education on SRCs), pairwise comparisons were restricted to gender and self-reported history of SRC; Wilcoxon rank-sum tests were performed to assess differences between these groups. A significance level of $p < 0.05$ was adopted for all analyses.

3. Results

3.1. Study participants

The online questionnaire was initiated by 4216 participants, and fully completed by 2582, translating to a completion rate of 61%. A total of 140 referees from Germany dropped out while completing the symptom checklist and were therefore not counted as 'full' completions; however, they provided answers to subsequent questions related to awareness and interest in receiving follow-up educational information.

Demographic data were available for a total of 2722 referees from Croatia (n = 30; 1%), Germany (n = 2522; 93%), Türkiye (n = 122; 4%), and Norway (n = 48; 2%); all data, stratified by country, are presented in Table 1. Overall, 179 referees (7%) were women, 2530 were men (93%), and 13 did not disclose gender (<1%). The median age was 29 years (IQR: 18–45). In terms of officiating level, 2342 referees (86%) reported refereeing at the amateur level, 488 (18%) at the regional level, and 239 (9%) at the highest level. Because multiple responses were allowed, percentages exceed 100%. Of all participants, 151 (~6%) reported prior formal training or education related to concussion management, while 2571 (94%) reported no prior training.

3.2. Sport-related concussion knowledge

The results of the binary knowledge questions and the symptom recognition are illustrated in Table 2. Median (IQR) CKS were 71% (61–85) in Croatia, 75% (61–86) in Germany, 71% (61–79) in Norway, and 71%

Table 1
Demographic characteristics of referees participating in the survey, stratified by country. Referee License refers to a certification issued by a national football association, authorizing an individual to officiate football matches. All data are shown as absolute and relative frequencies, n (%).

	Total n (%)	Croatia n (%)	Germany n (%)	Norway n (%)	Türkiye n (%)
N of referees	2722 (100)	30 (1,1)	2522 (92,6)	48 (1,8)	122 (4,5)
Gender					
Men	2530 (92,9)	29 (96,7)	2346 (93,0)	44 (91,7)	111 (91,0)
Women	179 (6,6)	1 (3,3)	163 (6,5)	4 (8,3)	11 (9,0)
Undisclosed	13 (0,5)		13 (0,5)		
Age					
Median (IQR)		29 (25–33)	28 (14–43)	25.5 (10–40)	35 (31–39)
14–18 yrs	641 (23,5)	0 (0)	623 (24,7)	18 (37,5)	0 (0)
19–25 yrs	452 (16,6)	1 (3,3)	444 (17,6)	6 (12,5)	1 (0,8)
26–30 yrs	255 (9,4)	17 (56,7)	215 (8,5)	4 (8,3)	19 (15,6)
31–35 yrs	251 (9,2)	3 (10,0)	203 (8,0)	2 (4,2)	43 (35,2)
>35 yrs	1068 (39,2)	9 (30,0)	982 (38,9)	18 (37,5)	59 (48,4)
Years of experience					
Median (IQR)		13 (11–19)	5 (2–15)	4 (2–11)	16 (12–20)
0–4 yrs	1149 (42,2)	0 (0)	1122 (44,5)	27 (56,3)	0 (0)
5–10 yrs	556 (20,4)	7 (23,3)	524 (20,8)	9 (18,8)	16 (13,1)
11–20 yrs	490 (18,0)	19 (63,3)	386 (15,3)	5 (10,4)	80 (65,6)
>20 yrs	484 (17,8)	4 (13,3)	447 (17,7)	7 (14,6)	26 (21,3)
Officiating level					
Amateur level	2342 (86,0)	1 (3,3)	2308 (91,5)	31 (64,6)	2 (1,6)
Regional level	488 (17,9)	7 (23,3)	443 (17,6)	28 (58,3)	10 (8,2)
Highest level	239 (8,7)	30 (100,0)	83 (3,3)	14 (29,2)	112 (91,8)
Age group					
U6-U12	996 (36,6)	9 (3,0)	965 (38,3)	6 (12,5)	16 (13,1)
U13-U19	2237 (82,2)	22 (73,3)	2150 (85,2)	42 (87,5)	23 (18,9)
U20-U23	2055 (75,5)	28 (93,3)	1865 (73,9)	41 (85,4)	121 (99,2)
Referee License	2679 (98,4)	30 (100,0)	2479 (98,3)	48 (100,0)	122 (100,0)
First Aid course	2361 (86,7)	28 (93,3)	2200 (87,2)	40 (83,3)	93 (76,2)
SRC education	151 (5,5)	8 (26,7)	108 (4,3)	1 (2,1)	34 (27,9)
History of SRC	628 (23,1)	7 (23,3)	608 (24,1)	8 (16,7)	5 (4,1)

Table 2
Correct responses (N) to concussion knowledge questions and symptom/sign recognition among referees stratified by country. Percentages indicate the proportion of participants in each country who correctly identified each question or symptom/sign. Items marked with an asterisk (*) denote SRC's symptoms/signs. Items without asterisk served as distractors. All data are shown as absolute and relative frequencies, n (%). For the German sample, denominators are reported for each item due to item-level dropouts; other country samples had complete responses.

Statement (correct response)	Total N = correct responses (%)	Croatia N = correct responses (%)	Germany N = correct responses (%)	Norway N = correct responses (%)	Türkiye N = correct responses (%)
Do you think that a loss of consciousness is always related to a sport-related concussion diagnosis? (FALSE)	2288 (84,1)	24 (80)	2123 (84)	46 (96)	95 (78)
Do you think that continuing to play with a sport-related concussion or symptoms of head injury leaves the young player vulnerable to greater injury? (TRUE)	2448 (89,9)	23 (77)	2261 (90)	45 (94)	119 (97)
After being knocked out during a match (or training session), a player can still participate in the same match (or training session) without any concern (FALSE)	2399 (88,1)	29 (97)	2210 (88)	46 (96)	114 (93)
Drowsiness*	1286 (48,9)	10 (33)	1129/2429 (46)	36 (75)	111 (91)
Nausea/Vomiting*	2436 (92,7)	29 (97)	2260/2427 (93)	43 (90)	104 (85)
Eating more than usual	2564 (97,6)	30 (100)	2366/2427 (97)	48 (100)	120 (98)
Dizziness/Balance problems*	2522 (96)	28 (93)	2337/2427 (96)	43 (90)	114 (93)
Hyperactivity/High energy	2528 (96,3)	29 (97)	2334/2425 (96)	48 (100)	117 (96)
Feeling of fainting*	1929 (73,4)	4 (13)	1808/2425 (74)	15 (31)	102 (84)
Excitement/Happiness	2576 (98,1)	30 (100)	2381/2425 (98)	47 (98)	118 (97)
Headache*	2511 (95,6)	29 (97)	2344/2425 (96)	43 (90)	95 (78)
Blurry vision/Double vision*	2300 (87,6)	23 (77)	2133/2424 (88)	36 (75)	108 (89)
Trouble paying attention*	1956 (74,5)	17 (57)	1827/2423 (75)	33 (69)	79 (65)
Talking more than usual	2362 (90,2)	29 (97)	2170/2418 (90)	47 (98)	116 (95)
Daydreaming more than usual*	440 (16,8)	4 (13)	383/2418 (16)	1 (2)	52 (43)
Difficulty concentrating*	2047 (78,1)	19 (63)	1921/2418 (79)	32 (67)	75 (61)
Difficulty remembering what people say*	1683 (64,4)	18 (60)	1573/2413 (65)	31 (65)	61 (50)
Calmness	2492 (92,9)	28 (93)	2326/2413 (96)	32 (67)	106 (87)
Being tired a lot/Get tired easily*	1180 (45,1)	13 (43)	1100/2413 (45)	37 (77)	30 (25)
Having troubles following directions*	1420 (54,3)	16 (53)	1329/2413 (55)	28 (58)	47 (39)
Elation	2573 (98,4)	30 (100)	2376/2413 (98)	48 (100)	119 (98)
Finding it hard to learn new things*	786 (30)	8 (27)	750/2413 (31)	7 (15)	21 (17)
Confusion*	2025 (77,4)	19 (63)	1916/2413 (79)	28 (58)	62 (51)
Neck pain or tenderness*	1619 (62,1)	14 (47)	1507/2407 (62)	21 (44)	77 (63)
Loss of consciousness*	1924 (73,8)	25 (83)	1755/2405 (72)	33 (69)	111 (91)
Numbness/Tingling in more than one arm or leg*	789 (30)	9 (30)	715/2394 (30)	8 (17)	57 (47)
Heart attack or heart abnormalities	2401 (92,9)	30 (100)	2221/2384 (93)	45 (94)	105 (86)
Irritability*	511 (19,7)	3 (10)	487/2382 (20)	10 (21)	11 (9)

(64–82) in Türkiye. The Kruskal-Wallis test showed no significant differences between countries ($\chi^2(3) = 3.32$, $p = 0.345$). The overall median CKS across all referees was 75% (61–86).

As shown in Table 2, the most recognized symptoms were dizziness (n = 2522; ≥96%) and headache (n = 2511; ≥95%), whereas irritability (n = 511; ≤20%) and numbness/tingling in more than one arm or leg (n = 789; 30%) were the least recognized. The distractor items were generally well recognized by the respondents, with ≥90% correct identification overall. However, two items fell below 90% correct identification in specific countries: calmness was correctly identified by 32 (67%) referees in Norway and 106 (87%) in Türkiye, and heart attack/heart abnormalities by 105 (86%) referees in Türkiye. Fig. 1 displays the radar plot of symptoms (distractors excluded) and the percentage of respondents who answered correctly, by country. Fig. S1, in the Supplements, represents the same, but for the distractor items.

3.3. Impact of gender and SRC history on concussion knowledge per country

In Germany, CKS percentage differed by gender and SRC history: women referees (n = 177) had significantly higher median scores [79% (68–89)] than men referees (n = 2346) [75% (61–86); $p < 0.001$]. Likewise, referees with a history of SRC (n = 608) scored higher [75% (64–86)] than those without a history of SRC (n = 1914) [75% (61–86); $p = 0.008$].

In Croatia, Norway, and Türkiye, gender comparisons were not calculated due to imbalanced groups (Croatia: 1/29 women/men; Norway: 4/44; Türkiye: 11/111). For SRC history, knowledge scores did not differ significantly in any country (Croatia: 7 yes/23 no, $p = 0.588$; Norway: 8/40, $p = 0.856$; Türkiye: 8/40, $p = 0.090$). Small numbers in these subgroups limit the interpretability of these results.

3.4. Awareness of any specific instrument to assess for SRC

Overall, 11% of participants (n = 289) reported being aware of a specific instrument to assess for SRCs. By country, awareness was 20% (n = 3) of the referees from Croatia, 12% from Germany (n = 263), 31% (n = 13) from Norway, and 11% (n = 10) from Türkiye. Only those who reported being aware were then asked, in an open-ended question, about the specific tools they were aware of. Among the respondents, the most frequently mentioned instrument was an oculomotor/eye-movement test (n = 260; 90% among those reporting awareness). In addition to oculomotor/eye movement-based tests, respondents also mentioned Maddocks questions and contacting medical personnel. Only few respondents explicitly cited the Sport Concussion Assessment Tool (SCAT; Croatia, n = 1; Germany, n = 5), which is not intended for immediate on-pitch assessment, and the Concussion Recognition Tool (CRT; Germany, n = 3), the latter being specifically designed for rapid on-field recognition in accordance with the Fédération Internationale de Football Association (FIFA) and UEFA guidelines.

3.5. Interest in receiving follow-up information

Overall, 1977 (73%) of referees expressed interest in receiving follow-up information. Preferred formats (multiple responses allowed) were presentations (48%), flyers (33%), websites (32%), webinars (25%), social media (31%), and books (9%). Regarding topics of interest (multiple responses possible), participants selected injury prevention (64%), risks associated with SRCs (55%), return-to-play (41%), and assessment of SRCs on the field (64%). Country-specific details are provided in the Supplements, Tables 1S and 2S.

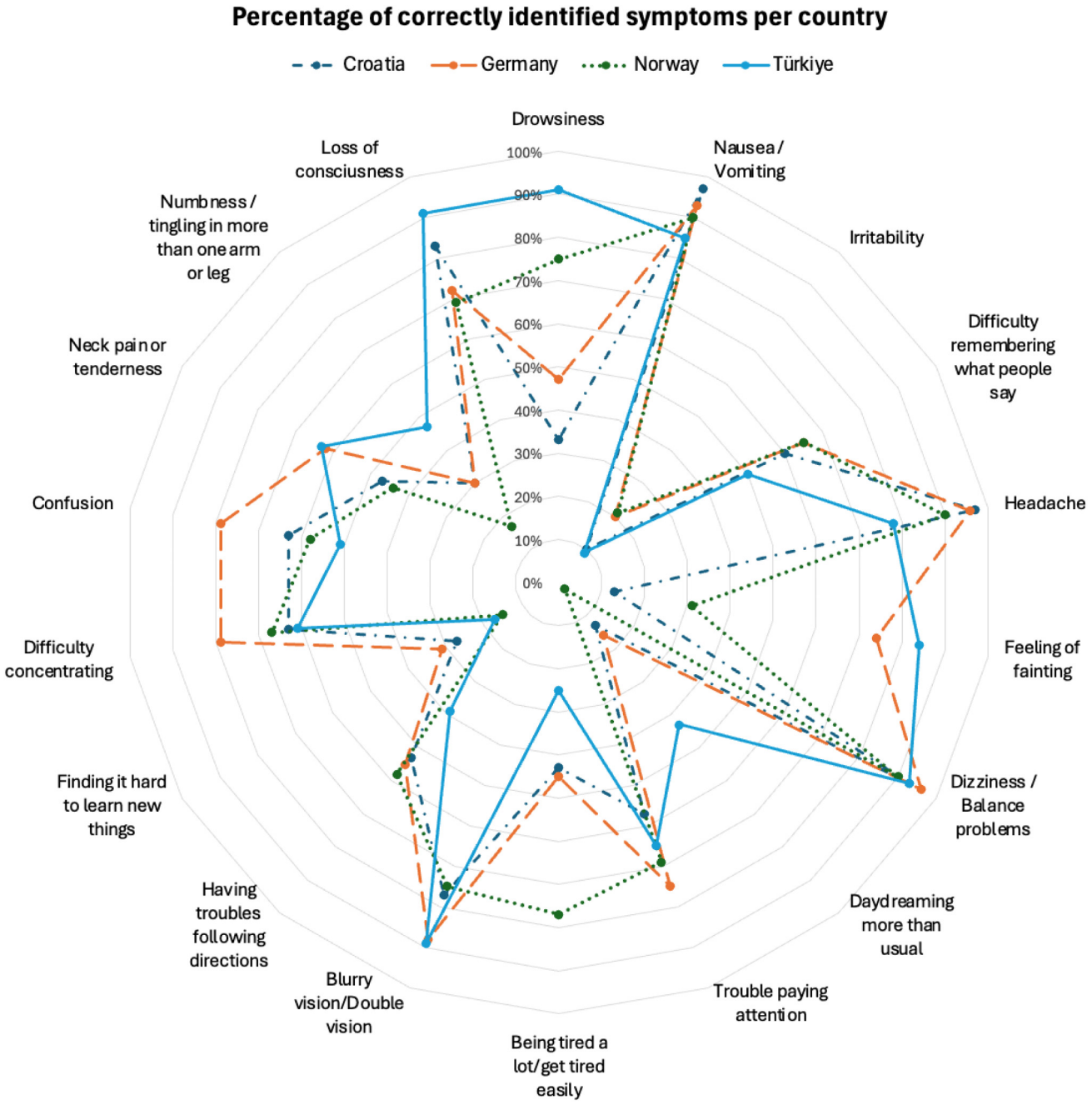


Fig. 1. Radar plot showing the percentage of correctly identified concussion-related symptoms per country. Each axis represents one symptom, and the distance from the center indicates the proportion of respondents correctly identifying that symptom. Lines represent each country: Croatia (blue dashed line), Germany (orange dashed line), Norway (green dotted line), and Türkiye (solid light blue line).

4. Discussion

Conducted within the European Union Erasmus + Sport project *REFORM*,[16] this study addresses a critical gap in player safety research by focusing on youth football referees. In this multi-country assessment of SRC knowledge and awareness of SRCs (Croatia, Germany, Norway, and Türkiye), referees demonstrated moderate knowledge, with no significant differences between countries, suggesting a broadly similar baseline across settings. In Germany, where statistical power was sufficient to test subgroups, women referees and those who suffered a prior SRC reported higher scores of CKS than men referees and those who did not suffer a prior SRC.

An important finding was the correct recognition of the pattern of SRC symptoms and distractors. Consistent with prior studies in football coaches and players, and referees in various youth sports,[14,17,18] referees in our study mostly correctly identified symptoms like headache, loss of consciousness, and dizziness, but showed lower recognition of less overt manifestations, such as irritability and increased daydreaming. While it may be easier to recognize subtle symptoms when one is familiar with the player, referees can still play a role by being aware of their significance and ensuring that players displaying abnormal behaviors are suspected of having an SRC and monitored appropriately. Although the overall median CKS was approximately 75%, this also indicates that around one-quarter of responses were incorrect across a large sample of referees. According to FIFA and UEFA concussion guidelines,[19,20] referees play a key role in the early recognition of potential SRCs by identifying visible signs and initiating the appropriate medical evaluation pathway. The observed knowledge gaps identified in this study,

particularly regarding symptom recognition and awareness of assessment tools, have direct implications for players' match safety in football. Insufficient or delayed recognition of subtle concussion signs such as irritability, numbness, or the feeling of fainting, may lead to continuing play rather than being stopped for medical evaluation. In consequence, players may get substituted later, potentially increasing the risk of detrimental effects on rehabilitation and player health due to continued physical effort, re-injury[21] and protracted recovery.[22] Prior research has identified referees as key stakeholders in injury prevention systems, due to their authority to stop play, remove athletes from the field, and initiate medical evaluation when a head injury is suspected.[13,14] Referees operate in a unique decision-making environment characterized by time pressure, limited visual angles, and the competing demand to maintain game flow, and concussion-specific knowledge may influence their ability to fulfill these responsibilities. Recognizing subtle behavioral or cognitive changes, understanding the implications of delayed symptom onset, and being aware of validated on-field tools are essential components of timely SRC identification. Thus, even minor improvements in knowledge or confidence may meaningfully shift real-time decisions, enabling timely concussion management or prevention.

This highlights the need to emphasize targeted education aligned with FIFA and UEFA concussion frameworks that cover the full spectrum of SRC symptoms, beyond the visible physical signs. In line with these protocols, on-pitch decision-making by referees should rely on recognition tools such as the CRT6, rather than multi-domain assessment tools like the SCAT6, which are designed for off-field clinical evaluation. Limited awareness of validated tools such as SCAT or CRT can be partially explained by the fact that, in our study, only ~5% of referees had ever received formal concussion education. Notably, 73% of referees expressed interest in receiving formal concussion education, indicating a high level of receptiveness to structured training initiatives within this population. Similar high interest was reported in prior studies. Rosenbloom et al.[8] reported high interest in future education among referees and King & Coughlan[14] showed that 87% of respondents thought that more could be done to educate youth sport referees, although in their study, the formal SRC education rate was much higher than in the present study (31% vs. 5%, respectively). Moreover, respondents frequently called for mandatory concussion recognition education as part of referee training.[14] This willingness to engage in further education not only reinforces the need for educational initiatives but also increases the feasibility and potential impact of implementing targeted, referee-specific concussion education programs. As such, embedding CRT6-based recognition content within referee training could enhance early identification of SRCs and support on-field decision-making, given that the CRT6 was specifically developed for non-medically trained individuals for the immediate recognition and management of suspected concussion.[23] In the German subgroup, where sample size permitted comparison, female referees demonstrated higher concussion knowledge scores than male referees. Prior research suggests that women demonstrate greater health-related vigilance and are more likely to engage with educational material on injury prevention.[24,25] This observation was confirmed by our findings, where German female referees showed a higher percentage of interest in educational material, compared to their male counterparts (80% vs. 70.5%, respectively). While this might partially explain the higher CKS observed among female referees in our sample, it does not indicate a causal relationship and should rather be seen as an association. No consistent subgroup differences were observed across other national samples, and imbalances in country-specific sample sizes precluded robust-stratified or multivariable analyses. In addition, findings for Germany should be interpreted cautiously and may not generalize to other countries due to the small sample sizes elsewhere.

Overall, the findings show variability in concussion knowledge among referees and underscore the importance of structured, referee-specific educational initiatives. Strengthening concussion knowledge may support the first step in the concussion management pathway, particularly in youth football settings where medical personnel are not routinely present. In this context, the present findings are currently being used as a basis for a need assessment to guide the development of referee-specific concussion education modules within the Erasmus + Sport REFORM project.[16]

4.1. Methodological considerations

This study is strengthened by its novelty, focus on referees in youth football, and its inclusion of referees from multiple European countries. However, there are several limitations to mention. First, interpretation should be made with caution due to a substantial imbalance in sample size between countries, with referees from Germany representing 93% of all respondents. This uneven distribution limits the representativeness of the findings and restricts the extent to which the study can be interpreted as a balanced multi-country comparison. We implemented the Kruskal-Wallis test as a non-parametric method to mitigate the impact of unequal group sizes. Yet, the imbalance may have affected the stability (i.e., reliability of the results if the sample composition changes) and sensitivity (i.e., ability of a test to detect real differences) of statistical comparisons. This directly increases the risk of both a type I error (detecting differences where none truly exist, driven by the dominant group) and a type II error (failing to detect true differences in smaller groups due to low statistical power). Thus, all between-country comparisons are interpreted as exploratory. Furthermore, the number of referees from Turkey, although higher, is unlikely to be representative of the national referee population given the country's size. It is important to mention that, as each national football association distributed the survey independently, the total number of referees who received the survey link was not available, and a response rate could not be calculated. Second, the cross-sectional design limits the ability to establish causal relationships. Third, there is no standardized survey for referees, and so far, no other study has been published assessing knowledge on SRCs in football referees only, limiting direct comparison. Fourth, the small number of female participants in some countries may have affected the generalizability of the findings. Additionally, analyses by officiating level, refereeing experience, and sociodemographic factors were not performed, as many referees reported officiating across multiple competitive levels, and country-specific subgroup sizes were insufficient to support stable-stratified comparisons. Fifth, the very low proportion of referees who had received concussion education restricted the ability to explore its relationship with knowledge. Finally, the use of a symptom checklist to calculate the CKS introduces the possibility of acquiescence bias, whereby individuals who tend to endorse all items may achieve higher scores regardless of actual knowledge. Although we included seven distractor symptoms aimed to partially mitigate this issue, checklist-based measures remain susceptible to overestimation of knowledge. In addition, the use of an available-case approach for calculating the CKS means that participants contributed data based on differing numbers of completed items, which may affect the comparability of scores. However, missing data were limited to the German sample and occurred only within the symptom-recognition checklist, suggesting that the overall impact on the findings is likely minimal. These limitations should be considered when interpreting the results.

This study assessed concussion-related knowledge but did not evaluate attitudes or behaviors, which represent subsequent stages in the behavior change framework.[15] While knowledge alone does not ensure safe practice, it is a prerequisite for recognizing potential SRCs and initiating appropriate action, particularly for referees acting under time pressure. Future research should extend this work by examining how knowledge translates into attitudes, confidence, and on-field decision-making behaviors.

5. Conclusion

This study demonstrates that football referees have a moderate level of SRC knowledge; while a substantial proportion of incorrect responses

highlight persistent knowledge gaps with potential implications for on-pitch decision-making. Gender-related differences were observed in the German sample, where subgroup size (men and women) was sufficient for analysis, whereas between-country comparisons were limited by sample imbalance. Given referees' responsibilities within FIFA and UEFA concussion protocols, and the fact that 73% expressed interest in receiving formal concussion education, these findings underscore the importance of targeted, role-specific concussion education aligned with current international guidelines.

CRediT authorship contribution statement

Giulia Gorgoni: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Resources, Data curation, Visualization, Writing – original draft, Writing – review & editing. **Gabriel Monthuley:** Conceptualization, Methodology, Validation, Data curation, Visualization, Writing – original draft, Writing – review & editing. **Thor Einar Andersen:** Conceptualization, Methodology, Validation, Writing – review & editing. **Nora Kümmeth-Gangel:** Conceptualization, Methodology, Validation, Investigation, Resources, Writing – original draft. **Ivana Pavlinac:** Conceptualization, Methodology, Resources, Writing – review & editing. **Zoran Dogas:** Conceptualization, Methodology, Resources, Writing – review & editing. **Anna Nordström:** Conceptualization, Methodology, Validation, Investigation, Resources, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Confirmation of ethical compliance

An ethics approval for this study was obtained from the Ethics Committee of Saarland University on 27/06/2024 (reference number: 24-21).

Funding information

This study was funded by the European Union Erasmus + Sport (grant number: 101133951).

Declaration of interest statement

The authors declare no conflicts of interest.

Acknowledgments

The authors thank all the referees for taking part in the study, and the national football associations of Croatia, Germany, Norway, and Türkiye for facilitating the survey distribution. Thank you also to all members of the REFORM working group who were involved in the project coordination and operational support. The authors gratefully acknowledge the Union of European Football Associations (UEFA) as an associated partner for their engagement, expertise, and long-standing commitment to player health and safety in football. UEFA contributed through contextual input and stakeholder dialog related to football practice.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jsams.2026.04.009.

References

1. Krutsch V, Krutsch W, Härtl J et al. Head injuries in professional football (soccer): results of video analysis verified by an accident insurance registry. *PLoS One* 2021;16(8). doi:10.1371/JOURNAL.PONE.0255695.
2. Mooney J, Self M, ReFaey K et al. Concussion in soccer: a comprehensive review of the literature. *Concussion (London, England)* 2020;5(3):CNC76. doi:10.2217/cnc-2020-0004.
3. Prien A, Grafe A, Rössler R et al. Epidemiology of head injuries focusing on concussions in team contact sports: a systematic review. *Sports Med* 2018;48(4):953-969. doi:10.1007/S40279-017-0854-4.
4. Patricios JS, Schneider KJ, Dvorak J et al. Consensus statement on concussion in sport: the 6th International Conference on Concussion in Sport–Amsterdam, October 2022. *Br J Sports Med* 2023;57(11):695. doi:10.1136/bjsports-2023-106898.
5. Feddermann-Demont N, Echemendia RJ, Schneider KJ et al. What domains of clinical function should be assessed after sport-related concussion? A systematic review. *Br J Sports Med* 2017;51(11):903. doi:10.1136/bjsports-2016-097403.
6. Mackay FD, Russel RE, Katy S et al. Neurodegenerative disease mortality among former professional soccer players. *N Engl J Med* 2019;381(19):1801-1808. doi:10.1056/NEJMoa1908483.
7. Ueda P, Pasternak B, Lim C-E et al. Neurodegenerative disease among male elite football (soccer) players in Sweden: a cohort study. *Lancet Public Health* 2023;8(4):e256-e265. doi:10.1016/S2468-2667(23)00027-0.
8. Rosenbloom C, Sean C, Ian B et al. 2.36 Concussion knowledge and experience amongst football referees in England. *Br J Sports Med* 2024;58(Suppl 1):A14. doi:10.1136/bjsports-2023-concussion.37.
9. Gouttebarge V, Cowie C, Goedhart E et al. Educational concussion module for professional footballers: from systematic development to feasibility and effect. *BMJ Open Sport Exerc Med* 2019;5(1). doi:10.1136/bmjsem-2018-000490.
10. FIFA. FIFA and World Health Organization launch global concussion awareness campaign. https://inside.fifa.com/health-and-medical/media-releases/fifa-and-world-health-organization-launch-global-concussion-awareness 2024.
11. FIFPRO. FIFPRO Europe and European Leagues join forces to improve concussion management in professional football. https://www.fifpro.org/news/fifpro-europe-and-european-leagues-join-forces-to-improve-concussion-management-in-professional-football/ 2022.
12. UEFA. UEFA launches concussion awareness campaign. UEFA.com: https://www.uefa.com/news-media/news/0256-0f8e7068a20e-9edc9f4af3a5-1000–uefa-launches-concussion-awareness-campaign/ 2019.
13. Yeo PC, Yeo EQY, Probert J et al. Systematic. review and qualitative analysis of concussion knowledge amongst sports coaches and match officials. *J Sports Sci Med* 2020;19. http://www.jssm.org.
14. King C, Coughlan E. Blowing the whistle on concussion knowledge and education in youth sport referees. *Open Access J Sports Med* 2021;12:109-117. doi:10.2147/OAJSM.S324191.
15. Rosenbaum AM, Arnett PA. The development of a survey to examine knowledge about and attitudes toward concussion in high-school students. *J Clin Exp Neuropsychol* 2010;32(1):44-55. doi:10.1080/13803390902806535.
16. Gorgoni G, Monthuley G, Andersen TE et al. Harmonisation and implementation of sports-related concussion guidelines in European youth football: the REFORM Erasmus+ project. *BMJ Open Sport Exerc Med* 2025;11(4). doi:10.1136/bmjsem-2025-002928.
17. Broglio SP, Vagnozzi R, Sabin M et al. Concussion occurrence and knowledge in Italian football (soccer). *J Sports Sci Med* 2010;9(3):418. [/pmc/articles/PMC3761699/].
18. Shafik A, Bennett P, Rosenbloom C et al. Sport-related concussion attitudes and knowledge in elite English female footballers. *Sci Med Footb* 2024;8(2):119-125. doi:10.1080/24733938.2022.2161613.
19. FIFA. *FIFA Concussion Protocol for Grassroots Football*, FIFA, 2025.
20. UEFA. *Concussion Procedures in UEFA Competitions*, UEFA Medical Regulations and Concussion Charter, 2024 https://documents.uefa.com/r/sulmvqJRvmKAZT4o7JV_jw/2dx46spFdFw4jqpGVGMo1A.
21. Nordström A, Nordström P, Ekstrand J. Sports-related concussion increases the risk of subsequent injury by about 50% in elite male football players. *Br J Sports Med* 2014;48(19):1447-1450. doi:10.1136/BJSPORTS-2013-093406.
22. Elbin RJ, Sufrinko A, Schatz P et al. Removal from play after concussion and recovery time. *Pediatrics* 2016;138(3):e20160910. doi:10.1542/peds.2016-0910.
23. Echemendia RJ, Ahmed OH, Bailey CM et al. The concussion recognition tool 6 (CRT6). *Br J Sports Med* 2023;57(11). doi:10.1136/bjsports-2023-107021.
24. Miyashita TL, Diakogeorgiou E, VanderVegt C. Gender differences in concussion reporting among high school athletes. *Sports Health* 2016;8(4):359-363. doi:10.1177/1941738116651856.
25. Sullivan L, Molcho M. Gender differences in concussion-related knowledge, attitudes and reporting-behaviours among high school student-athletes. *Int J Adolesc Med Health* 2021;33(1). doi:10.1515/ijamh-2018-0031.